



PhoenixUSBPro User Guide

Revision 4.0.6

2020-11-11

Revision History

Revision	Date	Author	Description
4.0.6	2020-11-11	AWA1695	1. Add Chapter 1: About This Document. 2. Update document template. 3. Update picture.
4.0.5	2020-09-21		Add the function descriptions about burning complete, unplugging USB, and clearing information.
4.0.4	2020-07-16		1. Add full-disk erase function. 2. Add secondary upgrade function.
4.0.2	2019-12-30		Use the template with watermark.
4.0.0	2014-08-01		



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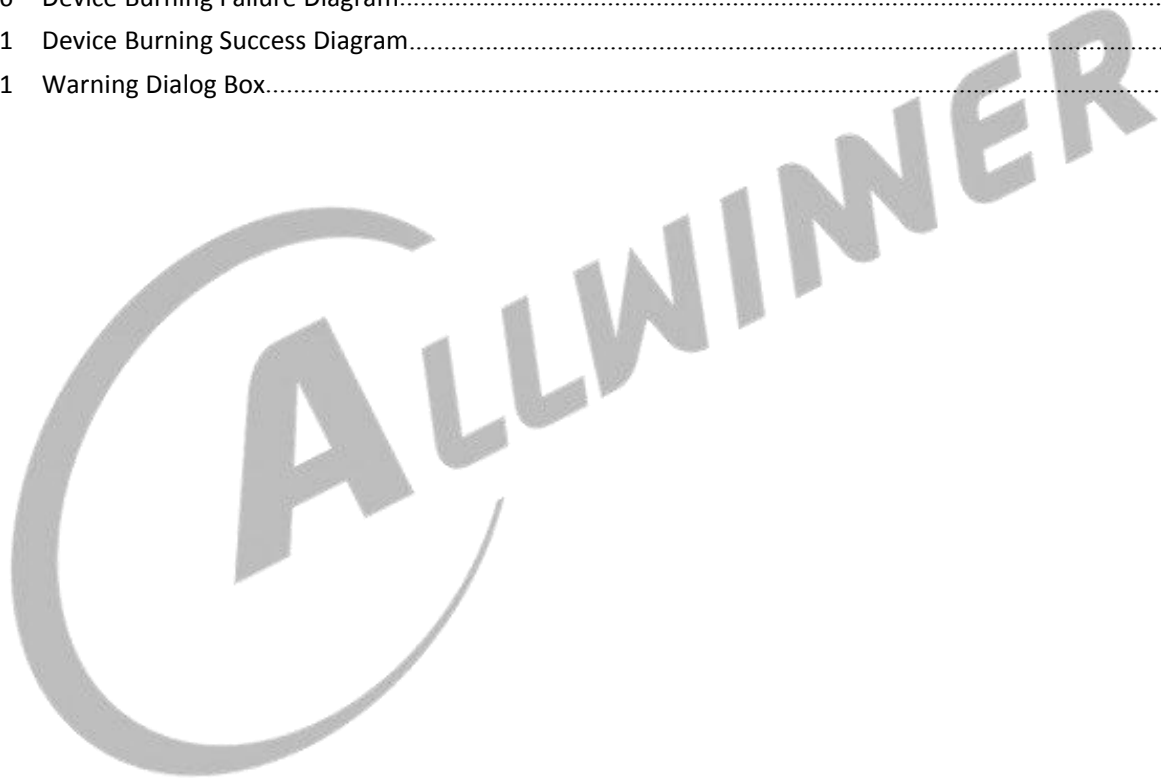
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1 About This Document

1.1 Purpose

This document introduces how to use PhoenixUSBPro.

1.2 Intended Audience

PhoenixUSBPro users

1.3 Related Versions

This document applies to the PhoenixUSBPro software running on Windows.

1.4 Conventions and Terminology

The document gives the following conventions.

1.4.1 Symbol Conventions

Use an eye-catching symbol such as “WARNING” or “CAUTION” or “NOTE” to immediately show users what to do to prevent accidents and keep a high level of safety. The definition is as follows.

Symbol	Description
 WARNING	Indicates potential risk of injury or death exists if the instructions are not obeyed.
 CAUTION	Indicates potential risk of equipment damage, data loss, performance degradation, or unexpected results exists if the instructions are not obeyed.
 NOTE	Provides additional information to emphasize or supplement important points of the main text.
 TIP	Indicates a tip that may help you solve a problem or save time.

2 Overview

2.1 Introduction to the PhoenixUSBPro

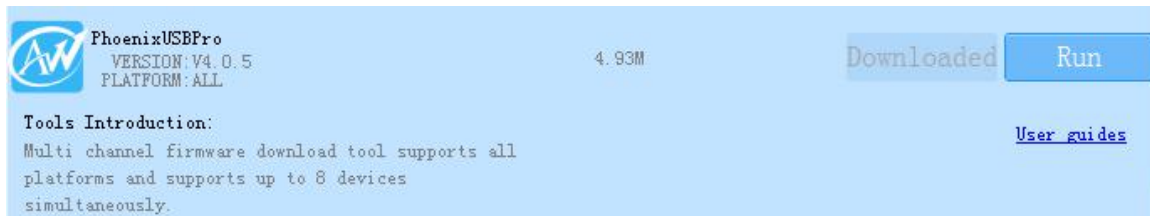
The PhoenixUSBPro is a multi-channel parallel firmware burning tool that supports up to eight devices to burn at the same time to improve mass production efficiency.

2.2 Running the Software

There are two ways to start the software.

(1). Find PhoenixUSBPro from Allwinner Product Support Tool (APST), click the “Run” button to run the software, as shown in Figure 2-1.

Figure 2- 1 APST Mode



(2). Run the executable file “PhoenixUSBPro.exe” in the tools folder, as shown in Figure 2-2.

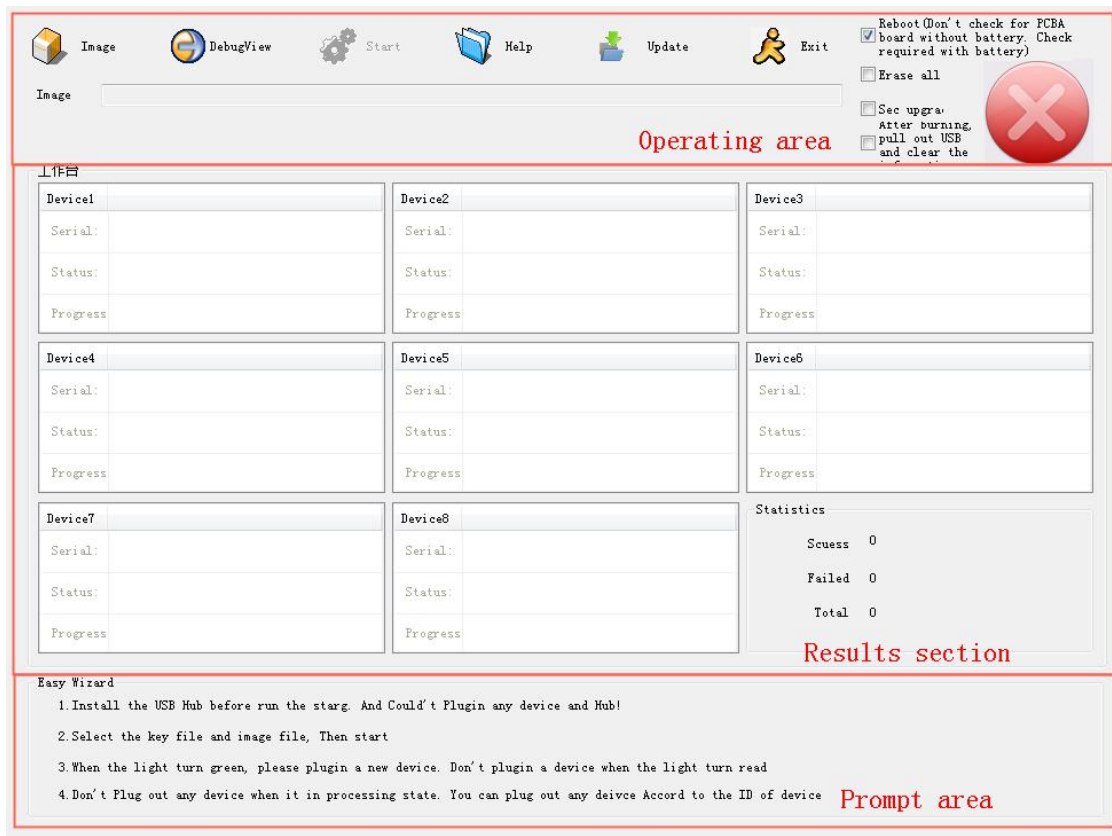
Figure 2- 2 Tools Folder Mode

	PhoenixUSBPro.exe	9/17/2020 10:34 AM	Application	652 KB
	UpdateVer.exe	4/6/2016 12:27 PM	Application	1,707 KB
	UpdateVerEx.exe	4/6/2016 12:27 PM	Application	1,707 KB
	AWCtrl.dll	5/29/2014 6:57 PM	Application extens...	124 KB
	AwPluginVector.dll	5/29/2014 6:57 PM	Application extens...	28 KB
	CommonFun.dll	5/29/2014 6:57 PM	Application extens...	105 KB

2.3 Software GUI

Figure 2-3 shows the GUI of PhoenixUSBPro.

Figure 2-3 Software GUI Diagram



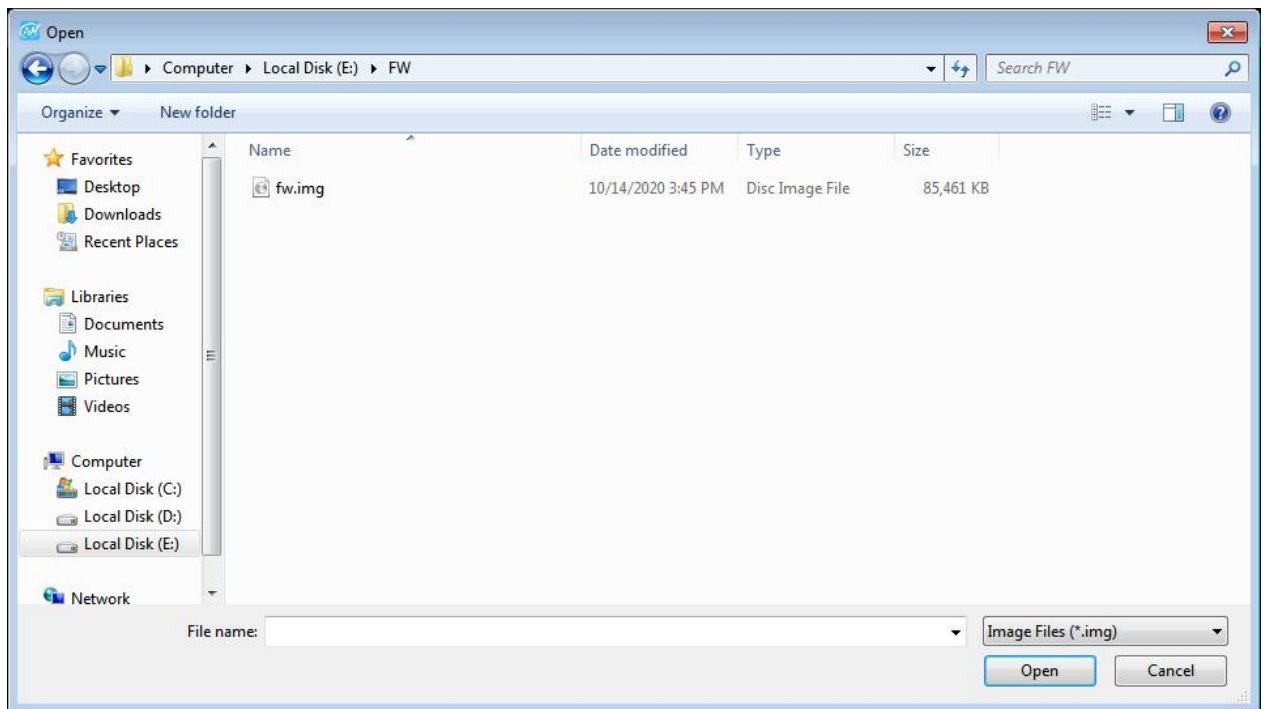
- 1) Operation area: Realize some functions such as selecting firmware, viewing debugging information, starting burning, opening user manual, checking version, exiting software, and selecting burning mode.
- 2) Result area: Display the burning process of the device, the burning result of the device, and the count of burning times since opening the tool.
- 3) Prompt area: Some important operation steps and precautions for using PhoenixUSBPro.

2.3.1 Operation Area

2.3.1.1 Selecting Firmware

Click  , and select the firmware file with the suffix *.img in the file selection dialog box that pops up, as shown in Figure 2-4.

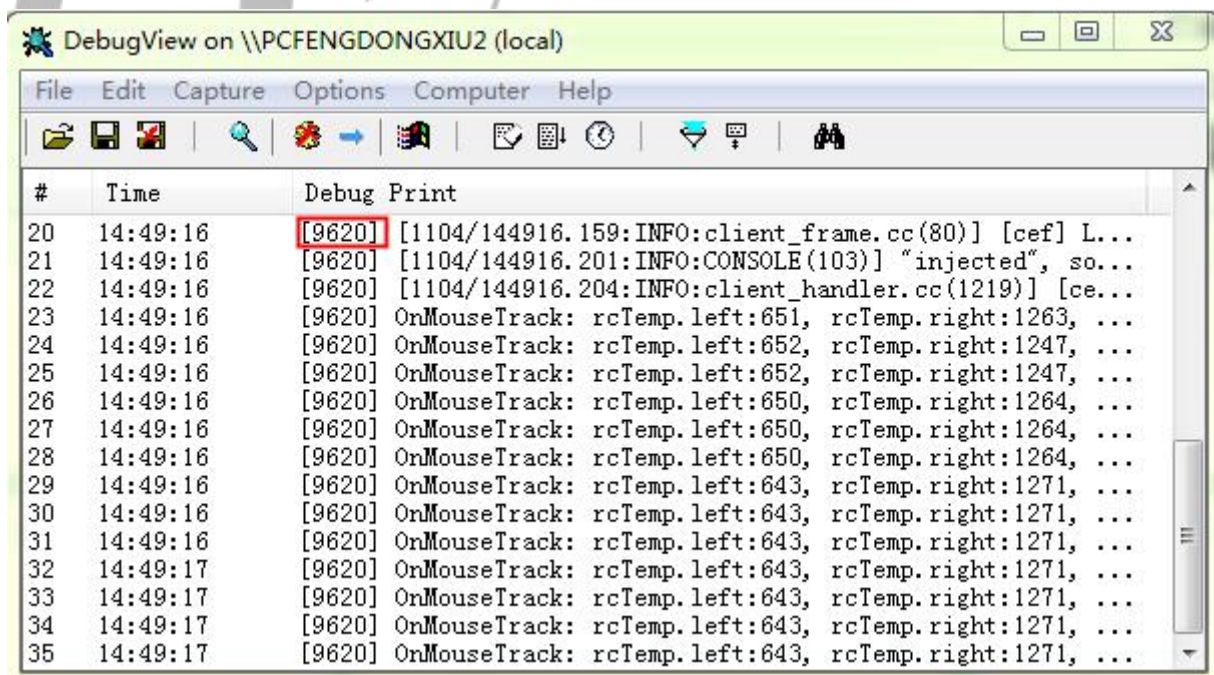
Figure 2- 4 File Selection Diagram



2.3.1.2 Viewing Debugging Formation

Setting the “debugging Information” button can view the printing information during burning and clicking this button can start the debugview.exe under PhoenixUSBPro installation directory, as shown in Figure 2-5. You can see the printing information of all Win32 programs in debugview. You can use its filtering function to just view the print information of the PhoenixUSBPro by filtering the process PID (view the process PID in the process page of the task manager, such as 9620 shown in Figure 2-5).

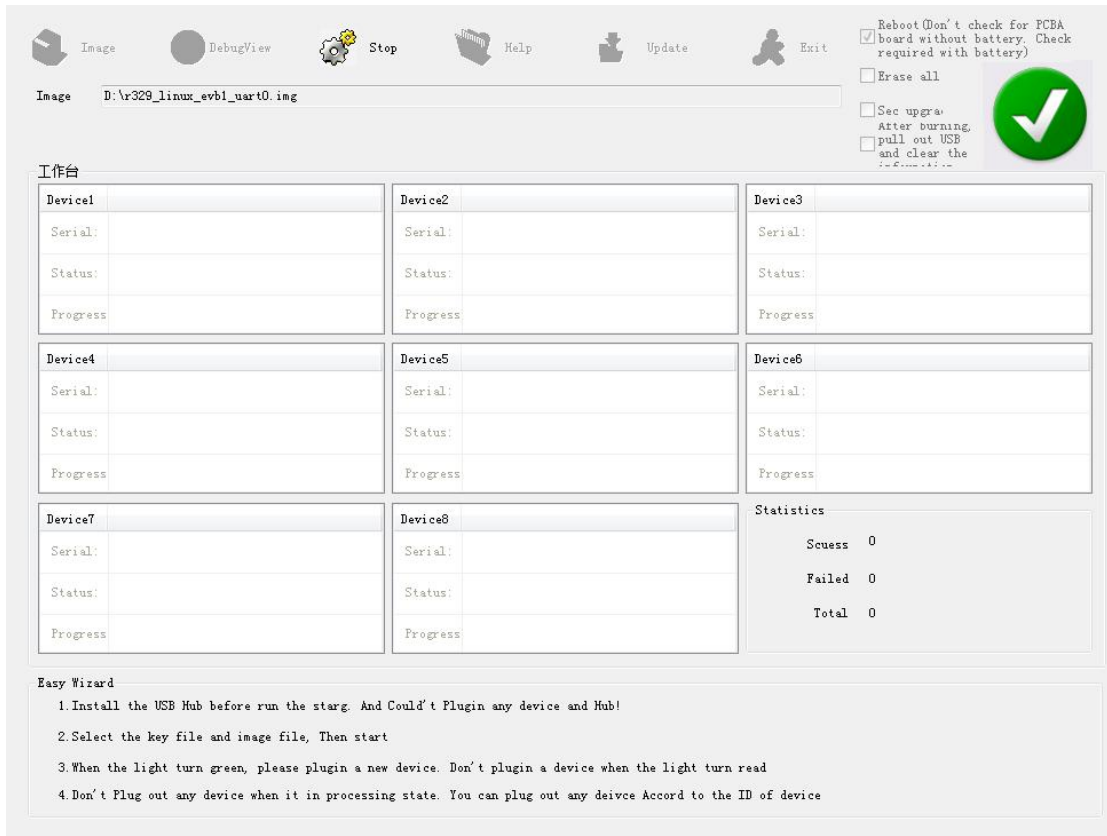
Figure 2- 5 Debugview GUI Diagram



2.3.1.3 Start

This button can only be used after the firmware is selected. Click , the user can see that the indicator light on the upper right of the main GUI is changed from  to . Meanwhile, other buttons are disabled, this button changes into , as shown in Figure 2-6.

Figure 2- 6 Starting Burning Diagram



1. Must unplug all devices before the mass-production software can be started.
2. When the indicator is , do not plug any devices.

2.3.1.4 Help

This button is to open this document.

2.3.1.5 Version Detection

This function is used to check whether the current tool is the latest version.

2.3.1.6 Exit

Exit the software.

2.3.1.7 Restart

The function is to restart the machine after burning.



Do not tick this button for empty board burning. Select this button for battery burning.

Reason: To prevent the machine connected to the battery from being in the uboot stage after burning, which will cause battery power to run out after a long time.

2.3.1.8 Full-Disk Erase

The function is to erase all contents in the entire flash memory (Flash), including the key area that has been burned.

2.3.1.9 Secondary Upgrade

The difference between this function and the normal burning process is that there is no need to press the FEL key or the combination key.

The secondary upgrade is easy to upgrade the upgraded machines.

Operation: Choose the right firmware, tick "secondary upgrade", click the "start" button, the machine can be mass-produced by plugging USB in the shutdown state.



1. This function is for the upgraded machine firmware that has been patched with uboot and turned on the auto-fel function.

2. Configure "CONFIG_CMD_SUNXI_AUTO_FEL=y" in uboot.

3. dts configuration:

[target]

auto_fel = 1

[platform]

sprite_keep_usb = 1

For the detailed uboot configuration above, please consult the engineer responsible for the uboot.

2.3.1.10 Clearing Information

The tool GUI provides this checkbox, which means that when this option is selected, the device information and burning status displayed in the corresponding area on the tool GUI will be cleared after the burning is

completed and the USB device is unplugged. The function is mainly used to ensure that the status information for the different devices can be updated in time during continuous burning and is not easy to cause confusion, which avoids the occurrence of missing burning. It can be ticked according to the actual requirements of users.

2.3.2 Result Area

The result area includes the device area and information area. The device area shows the specific burning process of each device and whether it has been successfully burned. The statistical information is the number of times for the successful burning, unsuccessful burning, and total burning after opening the statistical tool.

2.3.2.1 Device Area

It can be seen that PhoenixUSBPro can achieve the mass production of 8 devices at the same time, so when a device is plugged into a USB port of the HUB, then this USB port corresponds to the device number that appears on the tool GUI.

(1). Serial Number

This serial number is the device number assigned by the software to each USB port device. The serial number is randomly generated.

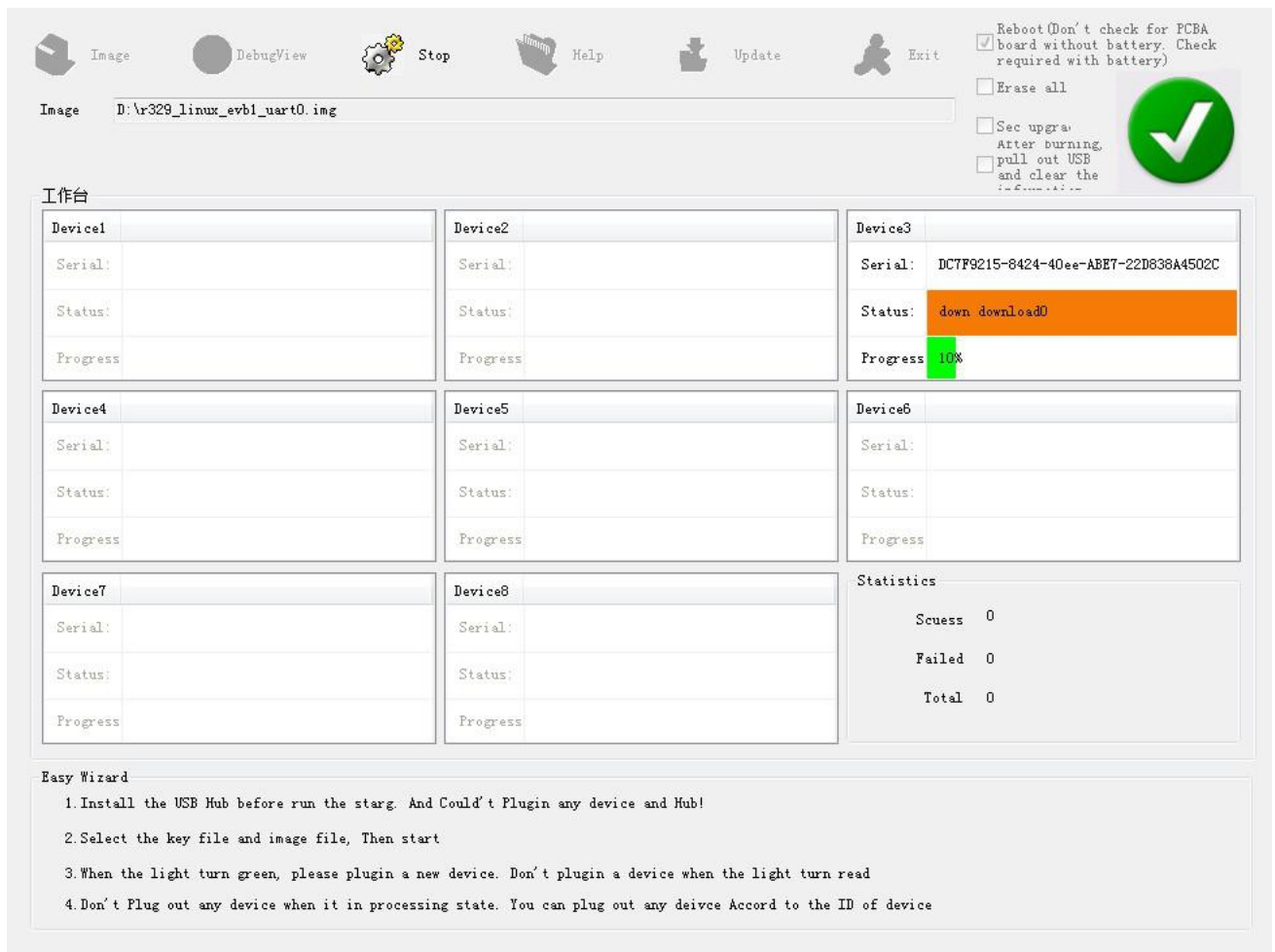
(2). Status

This bar shows the burning stage of the plugged device, such as downloading partition 0, as shown in Figure 2-7.

(3). Progress

This bar displays the burning progress. As shown in Figure 2-7.

Figure 2- 7 Device Burning Process Diagram



2.3.2.2 Statistics Information

The statistical information will be used to calculate and display the number of times for the successful burning, unsuccessful burning, and total burning after starting mass-production.

2.3.3 Prompt Area

The prompt area is mainly to remind users about some important operating steps and precautions during using PhoenixUSBPro.

3 Burning Device

3.1 USB Mass-Production

1.Check whether each USB port is working normally. If it cannot work properly, mark this port and do not use it during mass production.

2.Click the button, when the indicator light at the upper right of the main GUI changes into , plug the device into the USB port on your computer.



CAUTION

If the USB port is plugged into the device for the first time, a reminder to install the USB driver interface as shown in Figure 3-1 and Figure 3-2 may pop up. Please operate by following the selection of the red circle.

Figure 3- 1 Driver Installation Diagram

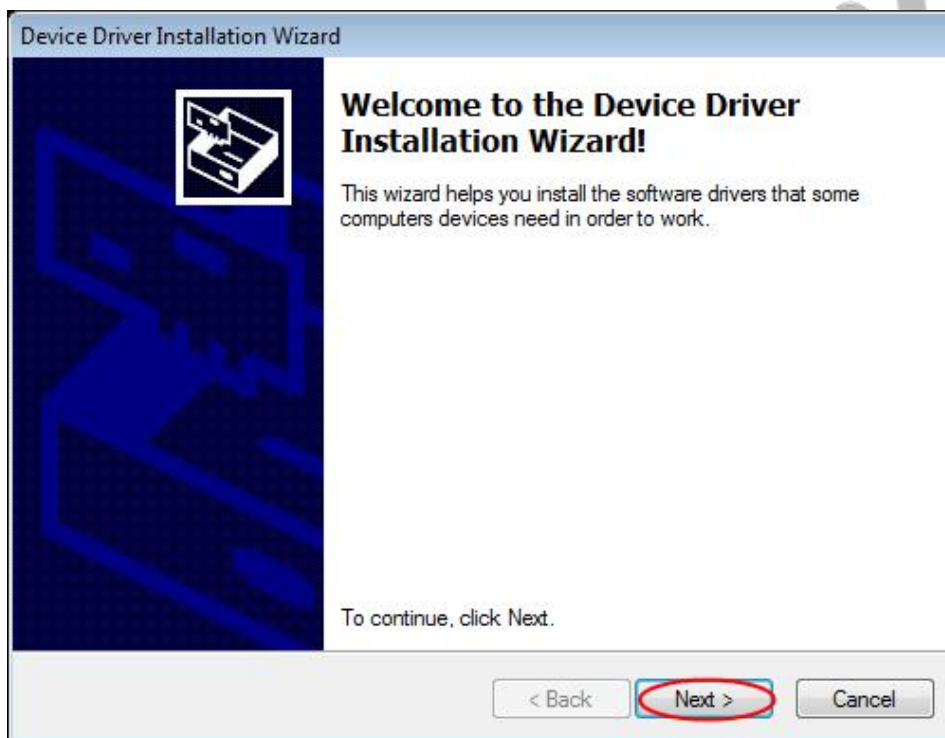


Figure 3- 2 Installation Completion Diagram



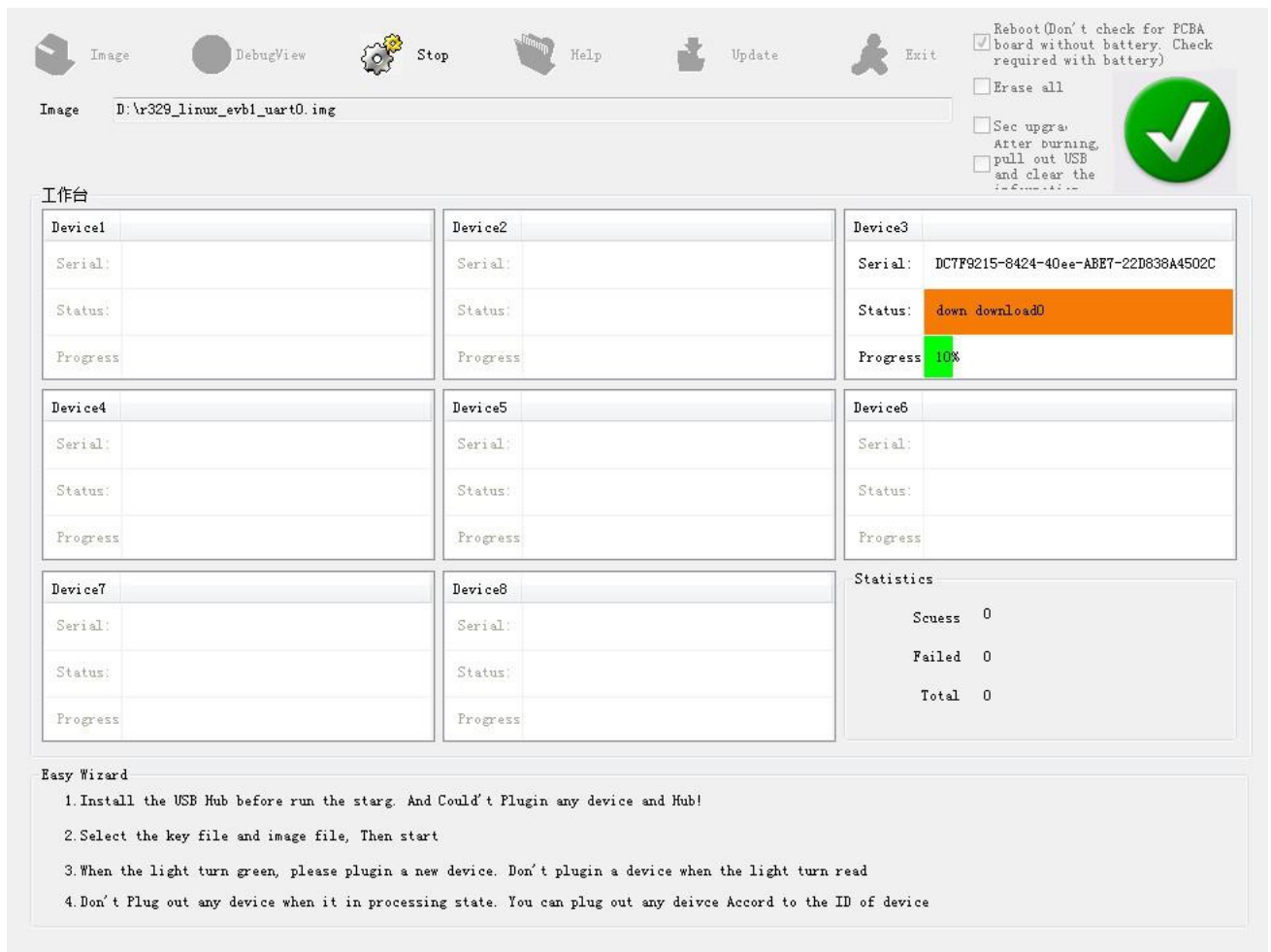
3.After inserting the device (as shown in Figure 3-3), the software will assign a device number to each USB port device. The USB port in Figure 3-3 corresponds to "Device 3", which means that the device ID of the USB port is 3.



CAUTION

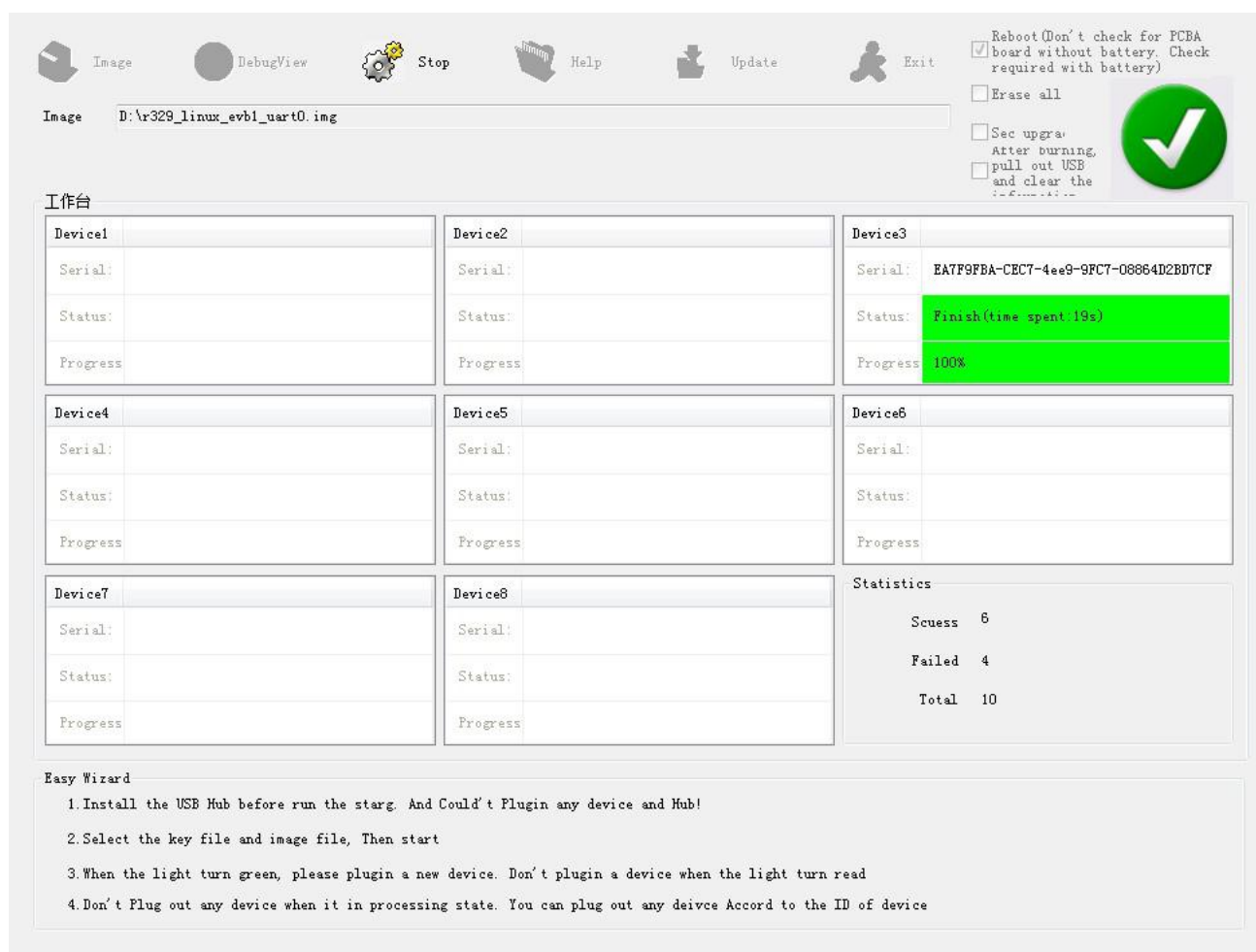
1. Please label the ID number of the corresponding USB cable (this only needs to be done once).
2. After the completion of mass production, unplug this device with the ID number of this label.

Figure 3-3 Device Burning Process Diagram



4. After the completion of mass production, the prompt as shown in Figure 3-4 will pop up. At the moment, you can unplug the device whose label ID is 3.

Figure 3- 4 Device Burning Completion Diagram



CAUTION

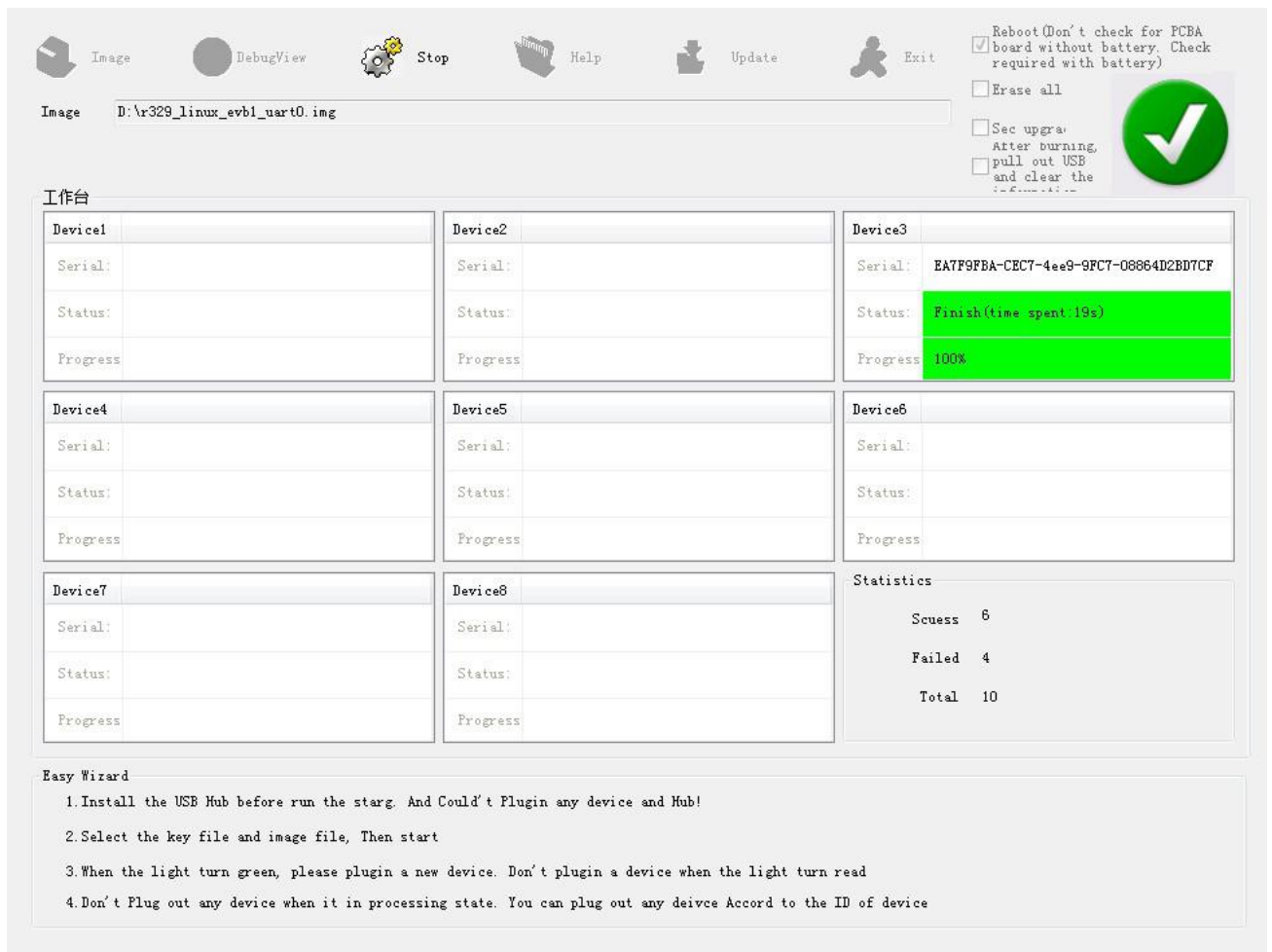
1. Only when the indicator is , the new device can be plugged.
2. After plugging the device, the indicator light stays  for 5 to 6 seconds before changing into .
3. If  is always present, it indicates the device has problems. See appendix A-FAQs of burning failure for the detail problems.

3.2 Judging the Status of Device Burning

3.2.1 Burning Success

When both the burning state and the burning progress are green, it means that the device is successfully burned, as shown in Figure 3-5, device 3 has been successfully burned.

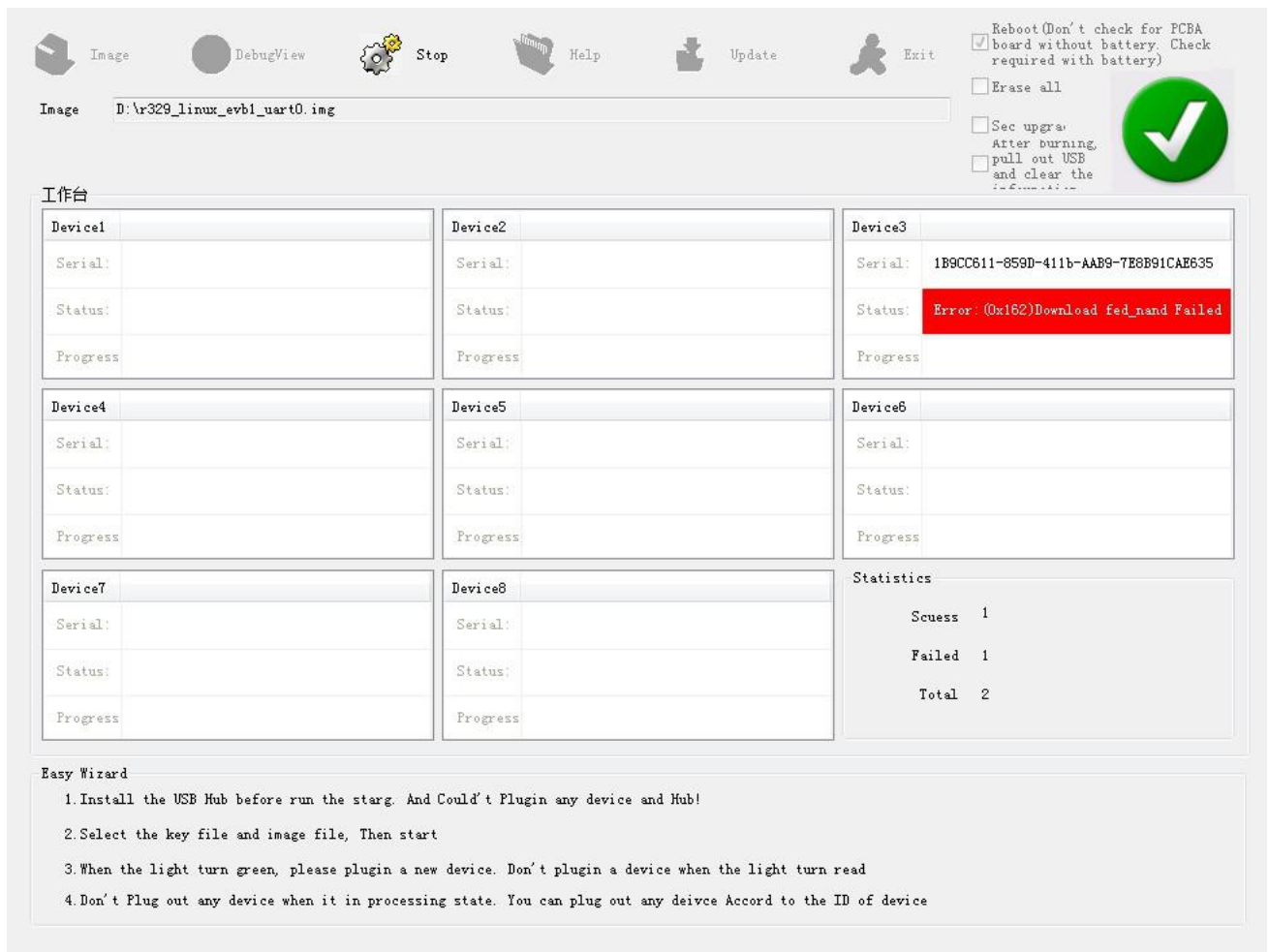
Figure 3- 5 Device Burning Successful Diagram



3.2.2 Burning Failure

When the burning status is red, it means that the device fails to burn, as shown in Figure 3-6.

Figure 3- 6 Device Burning Failure Diagram



4 Plugging and Unplugging Device

4.1 Plugging Device

After clicking , only when the indicator light is , a device can be inserted at this time.  indicates a new device cannot be plugged at this time, if a device is plugged at this time, the software will pop up a dialog box to warn.

4.2 Unplugging Device

Whether the device burns successfully or not will be displayed in the main GUI of the software. The user can unplug the corresponding device according to the device ID of the software. The result of burning will be indicated as success or failure by different colors.

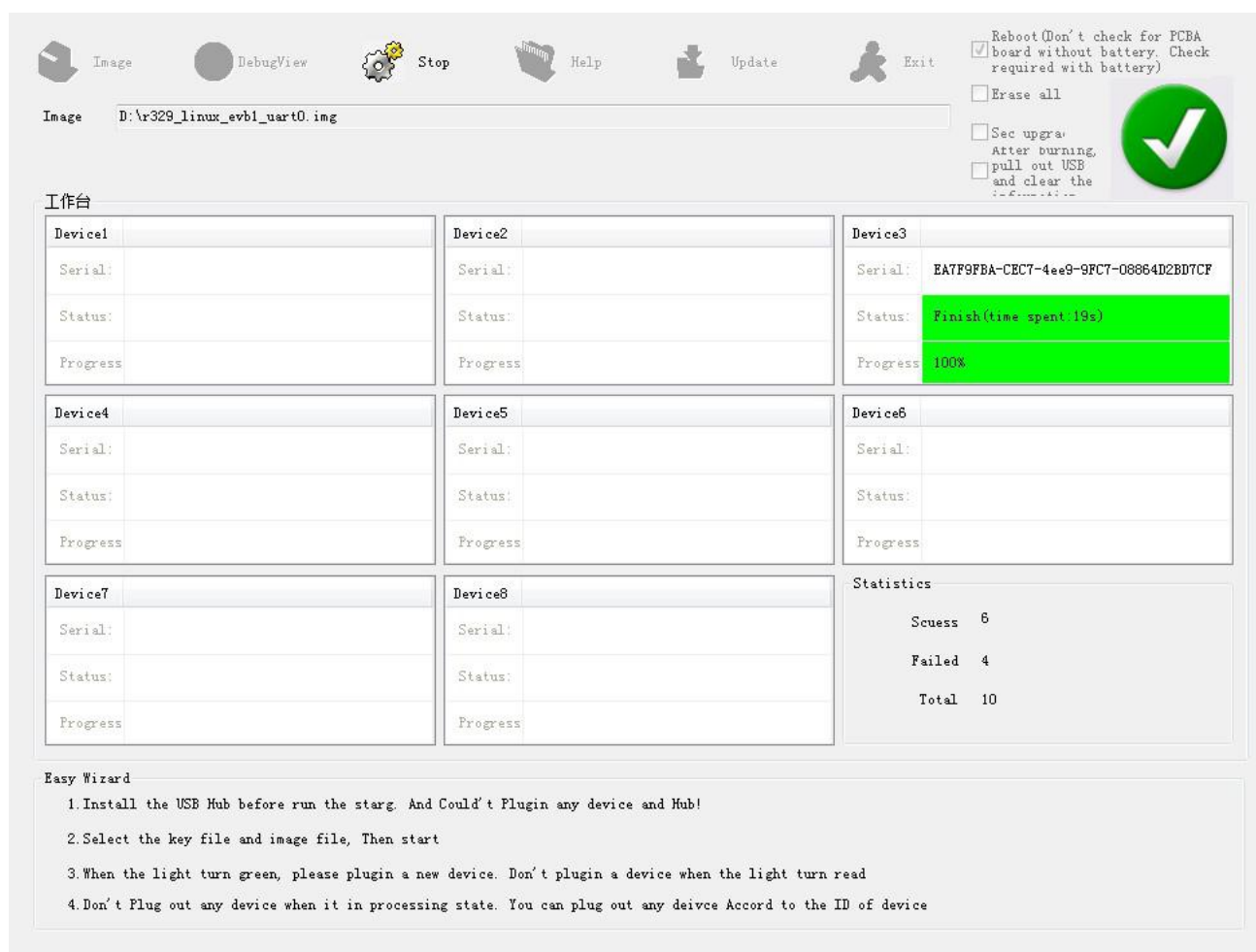
 The green indicates burning success.

 The red indicates burning failure. See the text description in the status bar for the failure reason.

 The orange indicates the burning is underway.

As shown in Figure 4-1, the device with ID 3 has been mass-produced, and users can unplug the device with ID 3.

Figure 4- 1 Device Burning Success Diagram



CAUTION

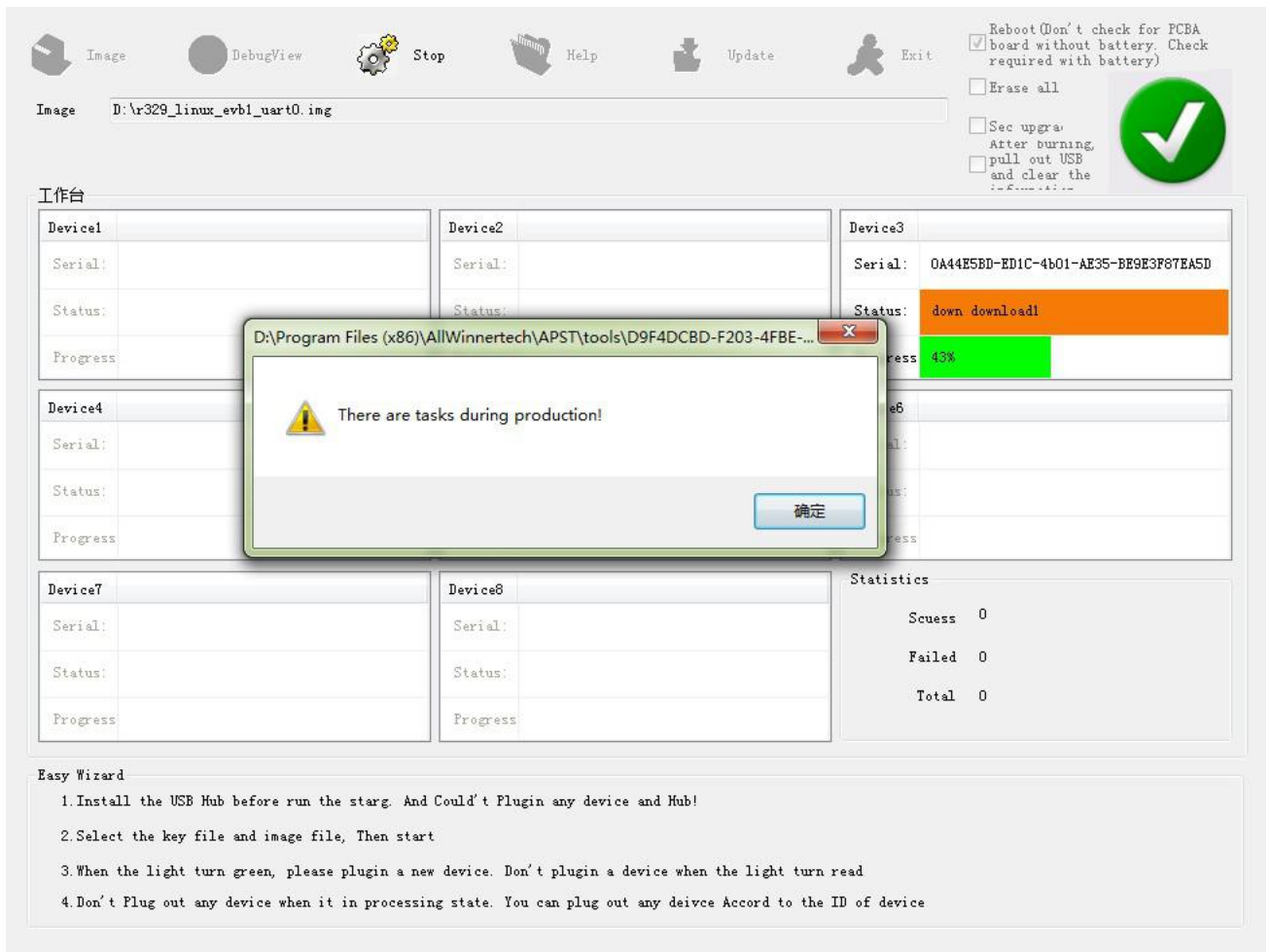
1. After opening the mass-production software, the HUB cannot be unplugged, otherwise, the system may crash.
2. During the process of device upgrading and burning, do not unplug the upgraded device. The device can be unplugged only after a prompt of success or failure appears on the device interface.

5 Stopping Mass-Production Software

1. Only when all devices are burned complete and the indicator light is , the software be effectively stopped.

2. When there are still devices that have not been burned, clicking  may pop-up a warning dialog box as shown in Figure 5-1, and the software cannot be stopped.

Figure 5- 1 Warning Dialog Box



Appendix A: FAQs of Burning Failure

A.1 Why does the indicator light is always after plugging a new device?

Cause Analysis:

1. The DRAM of the plugged device failed to initialize, so it cannot enter the upgrade state.
2. The plugged device has a battery and does not power off after the upgrade.

Solution:

1. Check the circuit to confirm whether the DRAM works normally.
2. Unplug the battery to make the device enter upgrade mode again, and plug the device into the computer.

A.2 How to deal with a prompt of reset after using more than 8 USB ports and changing USB port to plug a device?

Solution:

1. If you want to cancel the change, click "No" then unplug the device, otherwise go to step 2.
2. If other machines are burning, please wait for all machines to finish burning before selecting, or click "No" and wait for all machines to finish burning before plugging the device into the USB port.
3. Click "Yes" to re-mark the corresponding relationship between all USB interfaces and the device ID, and the USB interface of the plugged device will be marked as Device 1.
4. Re-select 8 USB ports and mark the corresponding device ID after plugging the device.

A.3 Why do some devices pop up automatically when you plug in a new device?

It may be caused by insufficient power supply due to too many devices connected via USB. Suggest that reduce the number of simultaneous mass-production devices and ensure enough USB power supply.

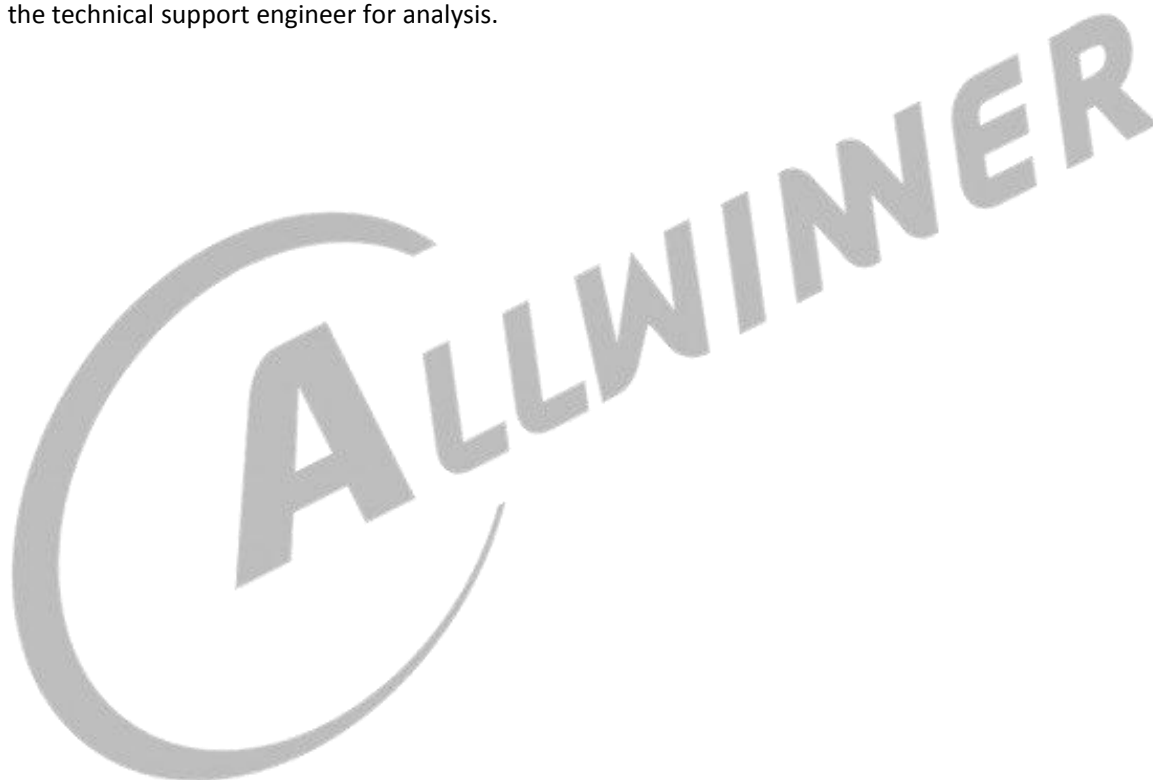
A.4 The common burning error number and description

Error Number	Error Description
0x100	System parameter loading exception
0x101	Open firmware package failure
0x102	Close firmware package failure
0x103	Open USB device failure
0x104	Close USB device failure
0x105	Build Master Boot Record (MBR) failure
0x120	Failed to scan storage media

Error Number	Error Description
0x160	Failed to download boot0
0x161	Failed to download boot1
0x162	Failed to download fes
0x163	Failed to burn mirror
0x167	Failed to download MBR partition
0x168	Failed to download complete
0x190	Failed to the system reset

A.5 The description of burning log

The log burned by the tool is in the logfile folder of the root directory of the tool. The number in the middle of the file name corresponds to the window on the interface, such as deciveid-6- logbuf.log Log of the sixth window on the corresponding interface. If you need technical support, please send the corresponding log to the technical support engineer for analysis.



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